

Software Requirements Specification (SRS)

Project: SkillSwap – Peer-to-Peer Skill Exchange Platform

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for SkillSwap, a web-based platform that enables students to exchange skills by offering tutoring services, requesting help, and scheduling sessions. This SRS serves as a reference for developers, testers, and stakeholders to ensure the system is implemented correctly and meets user needs.

1.2 Project Overview

SkillSwap is a full-stack web application that allows users to:

- Create accounts and manage profiles
- Offer skills through listings
- Request skills from others
- Book tutoring sessions
- Communicate through messaging
- Leave ratings and reviews
- Manage content through an admin panel

The system follows a client-server architecture using Django, MySQL, HTML/CSS, and JavaScript.

1.3 Scope

The SkillSwap platform will provide a centralized system for skill exchange between students. The platform will support listing creation, booking workflows, messaging, reviews, and administration tools.

Primary users:

- Learners (students seeking skills)
- Tutors (students offering skills)
- Administrators

The MVP focuses on essential marketplace and scheduling functionality.

1.4 Definitions

User: Any registered individual on SkillSwap

Listing: A skill offered by a tutor

Booking: A session request between learner and tutor

Skill Request: A post requesting help with a skill

Admin: System administrator managing platform

2. Overall Description

2.1 Product Perspective

SkillSwap is a standalone web application using:

Frontend: HTML, CSS, Django Templates

Backend: Django Framework

Database: MySQL

Authentication: Django Authentication System

2.2 System Architecture

Client → Django Backend → MySQL Database → Response to Client

2.3 User Classes

Learner:

- Search skills
- Book sessions
- Review tutors

Tutor:

- Create listings
- Accept or reject bookings
- Respond to skill requests

Admin:

- Manage users
 - Remove inappropriate content
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2.4 Assumptions

- Users have internet access
 - Users access system via web browser
 - Django and MySQL remain available
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2.5 Constraints

- System developed within 3 weeks
 - Uses Django and MySQL only
 - Web platform only
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3. Functional Requirements

Each requirement is uniquely numbered and testable.

3.1 User Authentication

FR-1: The system shall allow users to register with username, email, and password.

FR-2: The system shall allow users to login using valid credentials.

FR-3: The system shall allow users to logout.

FR-4: The system shall store passwords using secure hashing.

FR-5: The system shall allow users to edit profile information.

FR-6: The system shall display user profiles including skills and ratings.

3.2 Skill Listings

FR-7: The system shall allow tutors to create skill listings.

FR-8: The system shall allow tutors to edit listings.

FR-9: The system shall allow tutors to delete listings.

FR-10: The system shall allow users to view listings.

FR-11: The system shall display listing details including:

- Skill name
 - Description
 - Tutor name
 - Availability
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3.3 Search and Discovery

FR-12: The system shall allow users to search listings by keyword.

FR-13: The system shall allow users to browse all listings.

FR-14: The system shall allow filtering listings by category.

3.4 Booking System

FR-15: The system shall allow learners to request sessions.

FR-16: The system shall allow tutors to accept bookings.

FR-17: The system shall allow tutors to reject bookings.

FR-18: The system shall track booking status.

FR-19: The system shall display booking history.

FR-20: The system shall allow users to cancel bookings.

3.5 Messaging System

FR-21: The system shall allow messaging between users after booking acceptance.

FR-22: The system shall store message history.

FR-23: The system shall display message threads.

3.6 Rating and Review System

FR-24: The system shall allow learners to submit reviews.

FR-25: The system shall allow learners to rate tutors from 1 to 5.

FR-26: The system shall display average ratings.

3.7 Skill Request Board

FR-27: The system shall allow learners to create skill requests.

FR-28: The system shall allow tutors to view skill requests.

FR-29: The system shall allow tutors to apply to requests.

FR-30: The system shall allow learners to accept tutor applications.

FR-31: The system shall automatically create bookings after acceptance.

3.8 Administration

FR-32: The system shall allow admins to view users.

FR-33: The system shall allow admins to deactivate accounts.

FR-34: The system shall allow admins to delete listings.

4. Non-Functional Requirements

4.1 Performance

NFR-1: System shall respond within 2 seconds for all requests.

NFR-2: System shall support 100 concurrent users.

4.2 Security

NFR-3: Passwords shall be stored using hashing.

NFR-4: System shall restrict access using authentication.

4.3 Usability

NFR-5: System shall provide intuitive navigation.

NFR-6: Users shall complete booking in less than 5 steps.

4.4 Reliability

NFR-7: System uptime shall be 95 percent.

4.5 Maintainability

NFR-8: Code shall follow Django MVC structure.

5. Use Cases

Use Case 1: User Registration

Actor: User

Flow:

1. User clicks register
2. User enters information
3. System validates input
4. System creates account

Alternate Flow:

Invalid input → system shows error

Use Case 2: Create Listing

Actor: Tutor

Flow:

1. Tutor logs in

2. Tutor clicks create listing
 3. Tutor enters details
 4. System saves listing
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Use Case 3: Book Session

Actor: Learner

Flow:

1. Learner searches listing
 2. Learner clicks request booking
 3. Tutor accepts
 4. System confirms booking
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6. Out of Scope

The system will NOT include:

- Online payment processing
 - Video conferencing
 - Mobile application
 - AI recommendations
 - Email notifications
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7. Assumptions, Constraints, Dependencies

Assumptions:

- Users have browser access
- Django environment available

Constraints:

- Limited timeline
- Limited team size

Dependencies:

- Django framework
 - MySQL database
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8. System Models

Context Diagram Actors:

- Learner
- Tutor
- Admin

System:

SkillSwap Platform

Interactions:

- Login
 - Booking
 - Messaging
 - Reviews
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9. Database Model

Entities:

Users

Listings

Bookings

Messages

Reviews

SkillRequests

Applications

Relationships:

User → Listings

User → Bookings

Listing → Booking

User → Messages